

Lab 10 – Code Review and Quality: Using AI to Improve Code Quality and Readability

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Batch : 14

Task Description #1 – Syntax and Logic Errors

Prompt: Use AI to identify and fix syntax and logic errors in a faulty

Python script.

Input:

```
# Calculate average score of a student
```

```
def calc_average(marks):
```

```
    total = 0
```

```
    for m in marks:
```

```
        total += m
```

```
    average = total / len(marks)
```

```
    return average
```

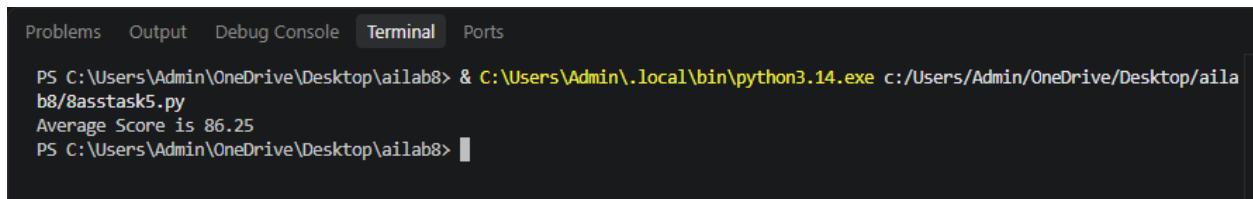
```
marks = [85, 90, 78, 92]
```

```
print("Average Score is", calc_average(marks))
```

Explanation:

The program calculates the total marks using a for loop. It divides the total marks by the number of marks to find the average. Fixed the indentation error, corrected avrage to average, and added the missing closing parenthesis.

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'Problems', 'Output', 'Debug Console', 'Terminal' (which is active), and 'Ports'. The terminal shows a PowerShell prompt 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>' followed by a command to run a Python script: '& C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py'. The output of the script is 'Average Score is 86.25'. The prompt returns to 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>' with a cursor at the end.

```
PS C:\Users\Admin\OneDrive\Desktop\ailab8> & C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py
Average Score is 86.25
PS C:\Users\Admin\OneDrive\Desktop\ailab8>
```

Task Description #2 – PEP 8 Compliance

Prompt: Use AI to refactor Python code to follow PEP 8 style guidelines.

Input:

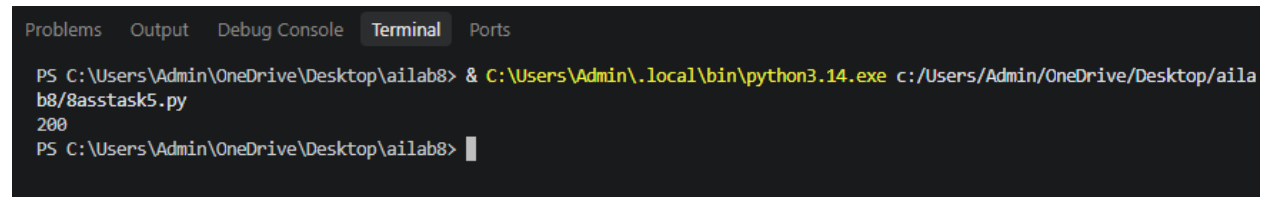
```
def area_of_rect(length, breadth):
    return length * breadth
```

```
print(area_of_rect(10, 20))
```

Explanation:

Added proper spaces and line breaks to make the code readable and follow PEP 8 guidelines. Changed L and B to length and breadth for meaningful variable names. The function multiplies length by breadth to calculate the rectangle's area.

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'Problems', 'Output', 'Debug Console', 'Terminal' (which is active), and 'Ports'. The terminal shows a PowerShell prompt 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>' followed by a command to run a Python script: '& C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py'. The output of the script is '200'. The prompt then returns to 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>' with a cursor at the end.

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Admin\OneDrive\Desktop\ailab8> & C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py
200
PS C:\Users\Admin\OneDrive\Desktop\ailab8> 
```

Task Description #3 – Readability Enhancement

Prompt: Use AI to make code more readable without changing its logic.

Input:

```
# Calculate percentage of a value
```

```
def calculate_percentage(value, percentage):
    return value * percentage / 100
```

```
total_value = 200
```

```
percentage_value = 15
```

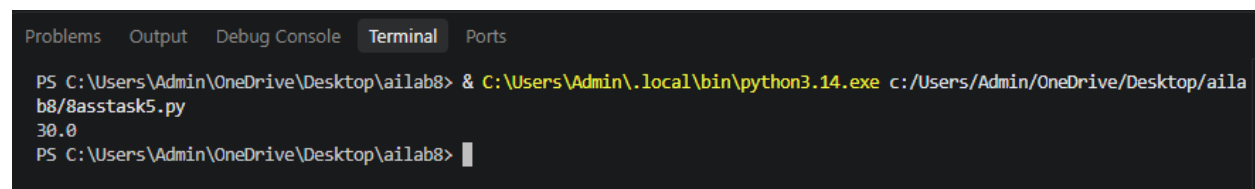
```
# Display the calculated result
```

```
print(calculate_percentage(total_value, percentage_value))
```

Explanation:

Changed variable names (c, x, y, a, b) to descriptive names for better readability. Added comments and proper indentation to make the code easy to understand. The function calculates 15% of 200 and displays the result without changing the original logic.

Output:

A screenshot of a terminal window with a dark background. The terminal shows a command prompt where a Python script is executed. The command is: `PS C:\Users\Admin\OneDrive\Desktop\ailab8> & C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py`. The output of the script is `30.0`. The prompt then returns to `PS C:\Users\Admin\OneDrive\Desktop\ailab8>`.

```
Problems  Output  Debug Console  Terminal  Ports
PS C:\Users\Admin\OneDrive\Desktop\ailab8> & C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py
30.0
PS C:\Users\Admin\OneDrive\Desktop\ailab8>
```

Task Description #4 – Refactoring for Maintainability

Prompt: Use AI to break repetitive or long code into reusable functions.

Input:

```
# Function to welcome a student
```

```
def welcome_student(name):
```

```
    print("Welcome", name)
```

```
students = ["Alice", "Bob", "Charlie"]
```

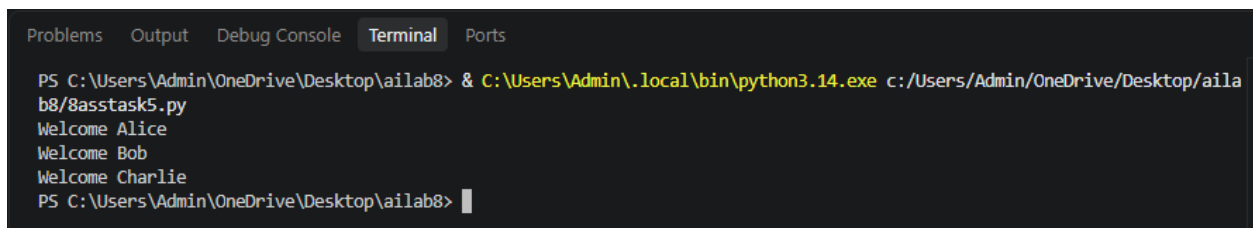
```
# Welcome all students using a loop
```

```
for student in students:  
    welcome_student(student)
```

Explanation:

Created a reusable function `welcome_student()` to display a welcome message for each student. Used a for loop to process all students instead of writing repetitive print statements. This makes the code shorter, easier to maintain, and reusable.

Output:

A screenshot of a Windows terminal window. The title bar shows 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The terminal content shows a PowerShell prompt 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>' followed by a command to run a Python script: '& C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py'. The output of the script is displayed as three lines: 'Welcome Alice', 'Welcome Bob', and 'Welcome Charlie'. The prompt returns to 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>'.

Task Description #5 – Performance Optimization

Prompt: Use AI to make the code run faster.

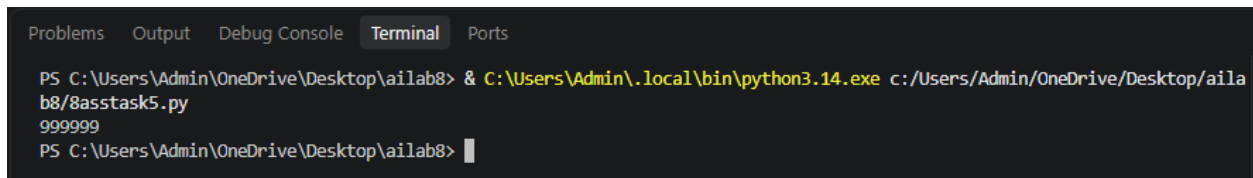
Input:

```
# Find squares of numbers using list comprehension  
squares = [n**2 for n in range(1, 1000000)]  
  
print(len(squares))
```

Explanation:

Used list comprehension to calculate squares in a single line instead of using a separate loop and append(). Removed the unnecessary nums list, reducing extra memory usage. The program calculates the squares of numbers from 1 to 999999 and prints the total count.

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'Problems', 'Output', 'Debug Console', 'Terminal' (which is active), and 'Ports'. The terminal shows a PowerShell prompt 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>' followed by the command '& C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py'. The output of the script is '999999'. The prompt continues with 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>' and a cursor.

```
Problems  Output  Debug Console  Terminal  Ports
PS C:\Users\Admin\OneDrive\Desktop\ailab8> & C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py
999999
PS C:\Users\Admin\OneDrive\Desktop\ailab8> |
```

Task Description #6 – Complexity Reduction

Prompt: Use AI to simplify overly complex logic.

Input:

```
def grade(score):
    if score >= 90:
        return "A"
    elif score >= 80:
        return "B"
    elif score >= 70:
        return "C"
    elif score >= 60:
        return "D"
    else:
```

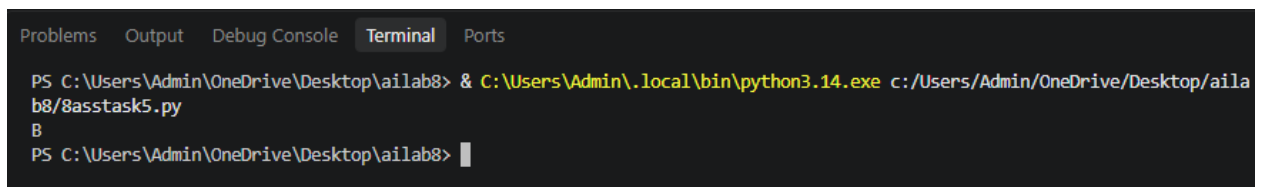
```
return "F"
```

```
print(grade(85))
```

Explanation:

Replaced nested if-else statements with elif to simplify the code. The function checks the score and returns the corresponding grade. This makes the code cleaner, easier to read, and easier to maintain.

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs for 'Problems', 'Output', 'Debug Console', 'Terminal' (which is active), and 'Ports'. The terminal shows a PowerShell prompt 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>' followed by a command to run a Python script: '& C:\Users\Admin\local\bin\python3.14.exe c:/Users/Admin/OneDrive/Desktop/ailab8/8asstask5.py'. The output of the script is 'B', which is displayed on the next line. The prompt returns to 'PS C:\Users\Admin\OneDrive\Desktop\ailab8>'.